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Diodes Inc. AP2112K-3.3TRG1 Datasheet

AP2112 Series 0.6 A 3.3 V Fixed Output SMT LDO Linear Regulator - SOT23-5

Absolute Maximum Ratings (Note 5)

Symbol	Parameter	Rating		Unit
V_{CC}	Power Supply Voltage	6.5		V
T_J	Operating Junction Temperature Range	+150		°C
T_{STG}	Storage Temperature Range	-65 to +150		°C
T_{LEAD}	Lead Temperature (Soldering, 10 Seconds)	+260		°C
θ_{JA}	Thermal Resistance (Junction to Ambient)(No Heatsink)	SOT25	184	°C/W
		SO-8	114	
		SOT89-5	120	
—	ESD (Machine Model)	400		V
—	ESD (Human Body Model)	4000		V

Note 5: Stresses greater than those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under "Recommended Operating Conditions" is not implied. Exposure to "Absolute Maximum Ratings" for extended periods may affect device reliability.

Recommended Operating Conditions

Symbol	Parameter	Min	Max	Unit
V_{IN}	Supply Voltage	2.5	6.0	V
T_A	Ambient Operation Temperature Range	-40	+85	°C

Electrical Characteristics

AP2112-1.2 Electrical Characteristics (@V_{IN} = 2.5V, C_{IN} = 1.0μF (Ceramic), C_{OUT} = 1.0μF (Ceramic), Typical T_A = +25°C, unless otherwise specified (Note 6))

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
V_{OUT}	Output Voltage	V_{IN} = 2.5V, 1mA ≤ I_{OUT} ≤ 30mA		V_{OUT} *98.5%	1.2	V_{OUT} *101.5%	V
$I_{OUT(MAX)}$	Maximum Output Current	V_{IN} = 2.5V, V_{OUT} = 1.182V to 1.218V		600	—	—	mA
$(\Delta V_{OUT}/V_{OUT})/\Delta I_{OUT}$	Load Regulation	V_{IN} = 2.5V, 1mA ≤ I_{OUT} ≤ 600mA		-1	0.2	1	%/A
$(\Delta V_{OUT}/V_{OUT})/\Delta V_{IN}$	Line Regulation	2.5V ≤ V_{IN} ≤ 6V, I_{OUT} = 30mA		-0.1	0.02	0.1	%/V
V_{DROP}	Dropout Voltage	I_{OUT} = 10mA		—	1000	1300	mV
		I_{OUT} = 300mA		—	1000	1300	
		I_{OUT} = 600mA		—	1000	1300	
I_Q	Quiescent Current	V_{IN} = 2.5V, I_{OUT} = 0mA		—	55	80	μA
I_{STD}	Standby Current	V_{IN} = 2.5V, V_{EN} in OFF mode		—	0.01	1.0	μA
PSRR	Power Supply Rejection Ratio	Ripple 0.5Vp-p V_{IN} = 2.5V, I_{OUT} = 100mA	f = 100Hz	—	65	—	dB
			f = 1kHz	—	65	—	
$(\Delta V_{OUT}/V_{OUT})/\Delta T$	Output Voltage Temperature Coefficient	I_{OUT} = 30mA T_A = -40°C to +85°C		—	±100	—	ppm/°C
I_{SHORT}	Short Current Limit	V_{OUT} = 0V		—	50	—	mA
V_{NOISE}	RMS Output Noise	No Load, 10Hz ≤ f ≤ 100kHz		—	50	—	μV_{RMS}
V_{IH}	V_{EN} High Voltage	Enable logic high, regulator on		1.5	—	6.0	V
V_{IL}	V_{EN} Low Voltage	Enable logic low, regulator off		0	—	0.4	
t_S	Start-up Time	No Load		—	20	—	μs
R_{PD}	EN Pull Down Resistor	—		—	3.0	—	MΩ
R_{DCHG}	V_{OUT} Discharge Resistor	Set EN pin at Low		—	60	—	Ω
T_{OTSD}	Thermal Shutdown Temperature	—		—	+160	—	°C
T_{HYOTSD}	Thermal Shutdown Hysteresis	—		—	+25	—	
θ_{JC}	Thermal Resistance (Junction to Case)	SOT25		—	96	—	°C/W
		SO-8		—	75	—	
		SOT89-5		—	47	—	

Note 6: Production testing at T_A = +25°C. Over temperature specifications guaranteed by design only.

Electrical Characteristics (Cont.)

AP2112-1.8 Electrical Characteristics (@V_{IN} = 2.8V, C_{IN} = 1.0μF (Ceramic), C_{OUT} = 1.0μF (Ceramic), Typical T_A = +25°C, unless otherwise specified (Note 6))

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
V_{OUT}	Output Voltage	V_{IN} = 2.8V, 1mA ≤ I_{OUT} ≤ 30mA		V_{OUT} *98.5%	1.8	V_{OUT} *101.5%	V
$I_{OUT(MAX)}$	Maximum Output Current	V_{IN} = 2.8V, V_{OUT} = 1.773V to 1.827V		600	—	—	mA
$(\Delta V_{OUT}/V_{OUT})/\Delta I_{OUT}$	Load Regulation	V_{OUT} = 1.8V, V_{IN} = V_{OUT}+1V, 1mA ≤ I_{OUT} ≤ 600mA		-1	0.2	1	%/A
$(\Delta V_{OUT}/V_{OUT})/\Delta V_{IN}$	Line Regulation	2.8V ≤ V_{IN} ≤ 6V, I_{OUT} = 30mA		-0.1	0.02	0.1	%/V
V_{DROP}	Dropout Voltage	I_{OUT} = 10mA		—	500	700	mV
		I_{OUT} = 300mA		—	500	700	
		I_{OUT} = 600mA		—	500	700	
I_Q	Quiescent Current	V_{IN} = 2.8V, I_{OUT} = 0mA		—	55	80	μA
I_{STD}	Standby Current	V_{IN} = 2.8V, V_{EN} in OFF mode		—	0.01	1.0	μA
PSRR	Power Supply Rejection Ratio	Ripple 0.5Vp-p V_{IN} = 2.8V, I_{OUT} = 100mA	f = 100Hz	—	65	—	dB
			f = 1kHz	—	65	—	
$(\Delta V_{OUT}/V_{OUT})/\Delta T$	Output Voltage Temperature Coefficient	I_{OUT} = 30mA T_A = -40°C to +85°C		—	±100	—	ppm/°C
I_{SHORT}	Short Current Limit	V_{OUT} = 0V		—	50	—	mA
V_{NOISE}	RMS Output Noise	No Load, 10Hz ≤ f ≤ 100kHz		—	50	—	μV_{RMS}
V_{IH}	V_{EN} High Voltage	Enable logic high, regulator on		1.5	—	6.0	V
V_{IL}	V_{EN} Low Voltage	Enable logic low, regulator off		0	—	0.4	
t_S	Start-up Time	No Load		—	20	—	μs
R_{PD}	EN Pull Down Resistor	—		—	3.0	—	MΩ
R_{DCHG}	V_{OUT} Discharge Resistor	Set EN pin at Low		—	60	—	Ω
T_{OTSD}	Thermal Shutdown Temperature	—		—	+160	—	°C
T_{HYOTSD}	Thermal Shutdown Hysteresis	—		—	+25	—	
θ_{JC}	Thermal Resistance (Junction to Case)	SOT25		—	96	—	°C/W
		SO-8		—	75	—	
		SOT89-5		—	47	—	

Note 6: Production testing at T_A = +25°C. Over temperature specifications guaranteed by design only.

Electrical Characteristics (Cont.)

AP2112-2.5 Electrical Characteristics (@V_{IN} = 3.5V, C_{IN} = 1.0μF (Ceramic), C_{OUT} = 1.0μF (Ceramic), Typical T_A = +25°C, unless otherwise specified (Note 6))

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
V_{OUT}	Output Voltage	V_{IN} = 3.5V, 1mA ≤ I_{OUT} ≤ 30mA		V_{OUT} *98.5%	2.5	V_{OUT} *101.5%	V
$I_{OUT(MAX)}$	Maximum Output Current	V_{IN} = 3.5V, V_{OUT} = 2.463V to 2.537V		600	—	—	mA
$(\Delta V_{OUT}/V_{OUT})/\Delta I_{OUT}$	Load Regulation	V_{OUT} = 2.5V, V_{IN} = V_{OUT}+1V, 1mA ≤ I_{OUT} ≤ 600mA		-1	0.2	1	%/A
$(\Delta V_{OUT}/V_{OUT})/\Delta V_{IN}$	Line Regulation	3.5V ≤ V_{IN} ≤ 6V, I_{OUT} = 30mA		-0.1	0.02	0.1	%/V
V_{DROP}	Dropout Voltage	I_{OUT} = 10mA		—	5	8	mV
		I_{OUT} = 300mA		—	125	200	
		I_{OUT} = 600mA		—	250	400	
I_Q	Quiescent Current	V_{IN} = 3.5V, I_{OUT} = 0mA		—	55	80	μA
I_{STD}	Standby Current	V_{IN} = 3.5V, V_{EN} in OFF mode		—	0.01	1.0	μA
PSRR	Power Supply Rejection Ratio	Ripple 0.5Vp-p V_{IN} = 3.5V, I_{OUT} = 100mA	f = 100Hz	—	65	—	dB
			f = 1KHz	—	65	—	
$(\Delta V_{OUT}/V_{OUT})/\Delta T$	Output Voltage Temperature Coefficient	I_{OUT} = 30mA T_A = -40°C to +85°C		—	±100	—	ppm/°C
I_{SHORT}	Short Current Limit	V_{OUT} = 0V		—	50	—	mA
V_{NOISE}	RMS Output Noise	No Load, 10Hz ≤ f ≤ 100kHz		—	50	—	μV_{RMS}
V_{IH}	V_{EN} High Voltage	Enable logic high, regulator on		1.5	—	6.0	V
V_{IL}	V_{EN} Low Voltage	Enable logic low, regulator off		0	—	0.4	
t_S	Start-up Time	No Load		—	20	—	μs
R_{PD}	EN Pull Down Resistor	—		—	3.0	—	MΩ
R_{DCHG}	V_{OUT} Discharge Resistor	Set EN pin at Low		—	60	—	Ω
T_{OTSD}	Thermal Shutdown Temperature	—		—	+160	—	°C
T_{HYOTSD}	Thermal Shutdown Hysteresis	—		—	+25	—	
θ_{JC}	Thermal Resistance (Junction to Case)	SOT25		—	96	—	°C/W
		SO-8		—	75	—	
		SOT89-5		—	47	—	

Note 6: Production testing at T_A = +25°C. Over temperature specifications guaranteed by design only.

Electrical Characteristics (Cont.)

AP2112-2.6 Electrical Characteristics (@V_{IN} = 3.6V, C_{IN} = 1.0μF (Ceramic), C_{OUT} = 1.0μF (Ceramic), Typical T_A = +25°C, unless otherwise specified (Note 6))

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
V_{OUT}	Output Voltage	V_{IN} = 3.6V, 1mA ≤ I_{OUT} ≤ 30mA		V_{OUT} *98.5%	2.6	V_{OUT} *101.5%	V
$I_{OUT(MAX)}$	Maximum Output Current	V_{IN} = 3.6V, V_{OUT} = 2.561V to 2.639V		600	—	—	mA
$(\Delta V_{OUT}/V_{OUT})/\Delta I_{OUT}$	Load Regulation	V_{OUT} = 2.6V, V_{IN} = V_{OUT}+1V, 1mA ≤ I_{OUT} ≤ 600mA		-1	0.2	1	%/A
$(\Delta V_{OUT}/V_{OUT})/\Delta V_{IN}$	Line Regulation	3.6V ≤ V_{IN} ≤ 6V, I_{OUT} = 30mA		-0.1	0.02	0.1	%/V
V_{DROP}	Dropout Voltage	I_{OUT} = 10mA		—	5	8	mV
		I_{OUT} = 300mA		—	125	200	
		I_{OUT} = 600mA		—	250	400	
I_Q	Quiescent Current	V_{IN} = 3.6V, I_{OUT} = 0mA		—	55	80	μA
I_{STD}	Standby Current	V_{IN} = 3.6V, V_{EN} in OFF mode		—	0.01	1.0	μA
PSRR	Power Supply Rejection Ratio	Ripple 0.5Vp-p V_{IN} = 3.6V, I_{OUT} = 100mA	f = 100Hz	—	65	—	dB
			f = 1kHz	—	65	—	
$(\Delta V_{OUT}/V_{OUT})/\Delta T$	Output Voltage Temperature Coefficient	I_{OUT} = 30mA T_A = -40°C to +85°C		—	±100	—	ppm/°C
I_{SHORT}	Short Current Limit	V_{OUT} = 0V		—	50	—	mA
V_{NOISE}	RMS Output Noise	No Load, 10Hz ≤ f ≤ 100kHz		—	50	—	μV_{RMS}
V_{IH}	V_{EN} High Voltage	Enable logic high, regulator on		1.5	—	6.0	V
V_{IL}	V_{EN} Low Voltage	Enable logic low, regulator off		0	—	0.4	
t_S	Start-up Time	No Load		—	20	—	μs
R_{PD}	EN Pull Down Resistor	—		—	3.0	—	MΩ
R_{DCHG}	V_{OUT} Discharge Resistor	Set EN pin at Low		—	60	—	Ω
T_{OTSD}	Thermal Shutdown Temperature	—		—	+160	—	°C
T_{HYOTSD}	Thermal Shutdown Hysteresis	—		—	+25	—	
θ_{JC}	Thermal Resistance (Junction to Case)	SOT25		—	96	—	°C/W
		SO-8		—	75	—	
		SOT89-5		—	47	—	

Note 6: Production testing at T_A = +25°C. Over temperature specifications guaranteed by design only.